Kye Son (Pyokyeong)

website: pyoky.me / me@pyoky.me
NYC –Tokyo – Seoul

Math & writing sample available at website: pyoky.me

DUKE UNIVERSITY

Bachelor of Science, Computer Science;
Bachelor of Science, Economics (Double Major);
Mathematics Minor; Foreign Lang. **Chinese** (2 yrs); **GPA: 3.82**/4.0; Selection of coursework: **CLASS OF '25**

- Math: *graduate* Mathematical Finance & Derivatives; *undergrad* Stat. Inference, Lin. Alg., Mult. Calc. *Durham, NC*
- CS: *graduate* Deep Learning, Algorithmic Game Theory; *undergrad.* Databases, Advanced Algorithms
- Economics: *graduate* Mathematical Finance & Derivatives; *undergrad.* Econometrics, Macro & Micro

WORK EXPERIENCE

*Currently at **Nomura Securities**, Risk Engine Developer*

- Large-scale **quantitative risk management (VaR, ES, etc.)** distributed data pipelines **PRESENT** NYC
- In **Python (Numpy, Dask, Prefect)**, Interfacing **fixed income** quant libraries (C++)
- **Claude AI** Skills to **automate** regression testing
- Exploring **Machine Learning** for Risk VaR Prediction (cost reduction)

Nomura Securities Market Risk Engines – *Intern* **SUMMER '24** NYC

- Groundbreaking division's LLM use in risk analysis through CoT reasoning, ReAct agent and tool-use.
- Automated fund report performance/quant extraction from reports using RAG and vision
- Wrote libraries interfacing LLM and OLAP/SQL DBs, automating risk analysis tasks

Hitachi Social Information Solutions – *Full-time Intern* **SUMMER '23** Tokyo

- Spearheaded administration front-end development (Angular) for Hitachi's Public Biometric Infrastructure initiative. Client was Japanese local government.
- Strengthened communication between developer and client with combined dev. and pros. mgmt. skills

Rep. of Korea Navy Interpretation Officer **NOV '20 - JUL '22**

- Lead interpretation for US-ROK Command Post Training (CCPT) exercises *Jinhae, South Korea*
- Managed visiting crew from U.K., U.S., Indonesia, Germany for technical submarine tours

Cydoc Medical Notes – *Front End & Graphic Design Intern* **SUMMER '20**

- Oversaw a front-end UI overhaul with Javacript, React and Semantic-UI, branding guidelines & logo; adopted for final launch product *Durham, NC*

KKMani's Sprout Team – *Financial News on Renewable Energy* **SUMMER '19** Tokyo

- Full rewrite of a cross-platform email framework with HTML & CSS,

RESEARCH

*Currently investigating **Ethereum Restaking (Eigenlayer) protocol** cryptoanalytic security* **PRESENT**

- with Duke PhD candates in algorithmic game theory, theory of computation

Wolfram Research Summer Program – "*Exploration of Fundamental Maths via Implementation of Common Axiom Systems and Proof Generation*" (link) **SUMMER '18** Boston

PROJECTS

Homelab project with **Debian/Arch Linux**, ESXi & Proxmox, with second-hand server parts
Maintained services such as local DNS, OpenVPN, Apache web server for personal website
Contribution to **rust-coreutils** as part of OSS contribution and personal interest in Rust language design

SKILLS

Proficient in **Python** (ML in PyTorch, Data Analysis in Prefect, Pandas), **MinIO**, **Nexus**, **CI/CD**, Advanced **SQL**, Apache **Avro**, **Parquet** formats, **Linux Internals, C & ARM Assembly** (Coursework), **Mathematica** (Math Research), **Javascript/Typsscript** with **React & Angular** (Work experience)

LANGUAGES

Fluent in **English** (*TOEFL:* 119/120, *IELTS:* 8.5/9), **Japanese** (*JLPT N-1:* 175/180)**, Korean** (*Native Speaker*)
Business capable in **Chinese** (2 years coursework)

LEADERSHIP

Lambda Phi Epsilon, Duke Associate Chapter, **Treasurer** (2yrs.)

- Orchestrated 20-year chapter alumni anniversary event with Hilton Hotel and Duke univ. for largest Asian-American fraternity in the U.S.

TAZNs Acapella Group, VP (1yr.), **Treasurer** (1yr.)

- Spearheaded group revitalization post-Covid–19, leading events, performances and communication